

a Training-to-design paradigm

Pre-training

Single-sequence genome foundation model

Diverse genomes (single-sequence training)



Objective: next-token likelihood

A T G C T C G T ... [A]

Context length (tokens)



Model scale (parameters)



Post-training

Evolution-informed post-training

MSA context

... 120 121 122 123 124 125 126 ...

sp.1 ... A C G T C G A ...

sp.2 ... A T G T T T A ...

... ..

target ... A G T C C T A ...

training only; no MSA at inference



Variant-effect scoring (Δ log-likelihood, ref $\rightarrow$ alt)

Reference A G T C A C T ...

Mutant A G T T A C T ...



Variant effect

Pathogenicity

Functional impact

Reinforcement learning

